INVENTORY AND SUPPLIER MANAGEMENT SYSTEM

Presented By : Wai Yan Lynn 67111411
Min Hein Thu 67111391
Pyae Sone 67111336

SYSTEM OVEVIEW

- Manage stock levels and item movement transactions.
- Create, approve, issue and receive supplier orders.
- Generate low stock alerts when item quantity reaches the threshold.
- Produce operational and management reports.
- Control user accounts, roles and access rights.

SCOPE DEFINITION

IN SCOPE

- Stock Tracking
- Supplier Orders
- Low Stock Alerts
- Reporting
- User Management
- Audit Logs

OUT OF SCOPE

- Finance and accounting
- Logistics and delivery routing
- Supplier Portal
- POS Integration

SYSTEM MODULES

STOCK TRACKING

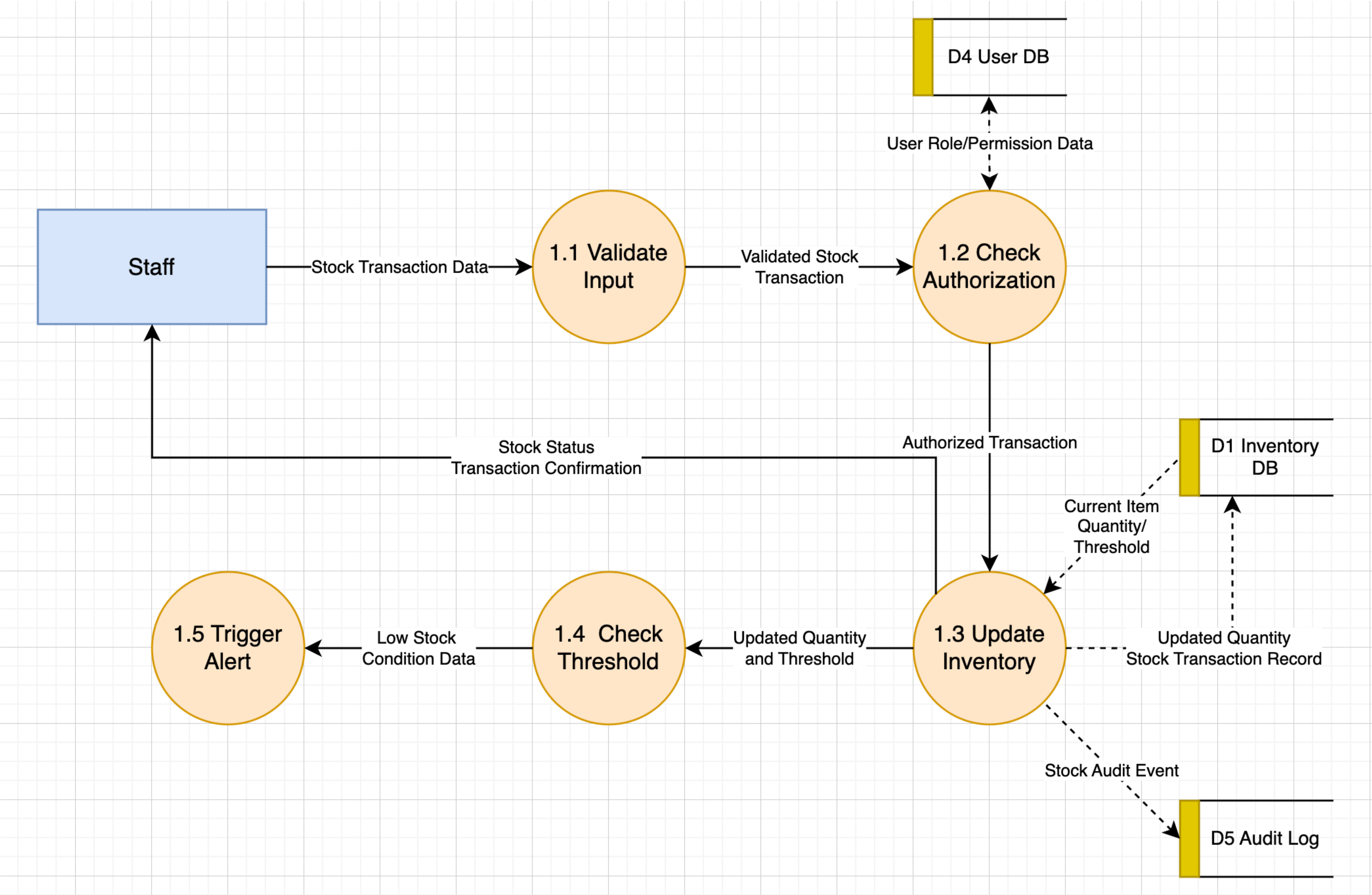
SUPPLIER ORDER

LOW STOCK ALERT

USER MANAGEMENT

REPORTING

1.0 STOCK TRACKING MODULE



FUNCTIONAL REQUIREMENTS

- The system shall create a low stock alert when the current quantity is less than or equal to the threshold value.

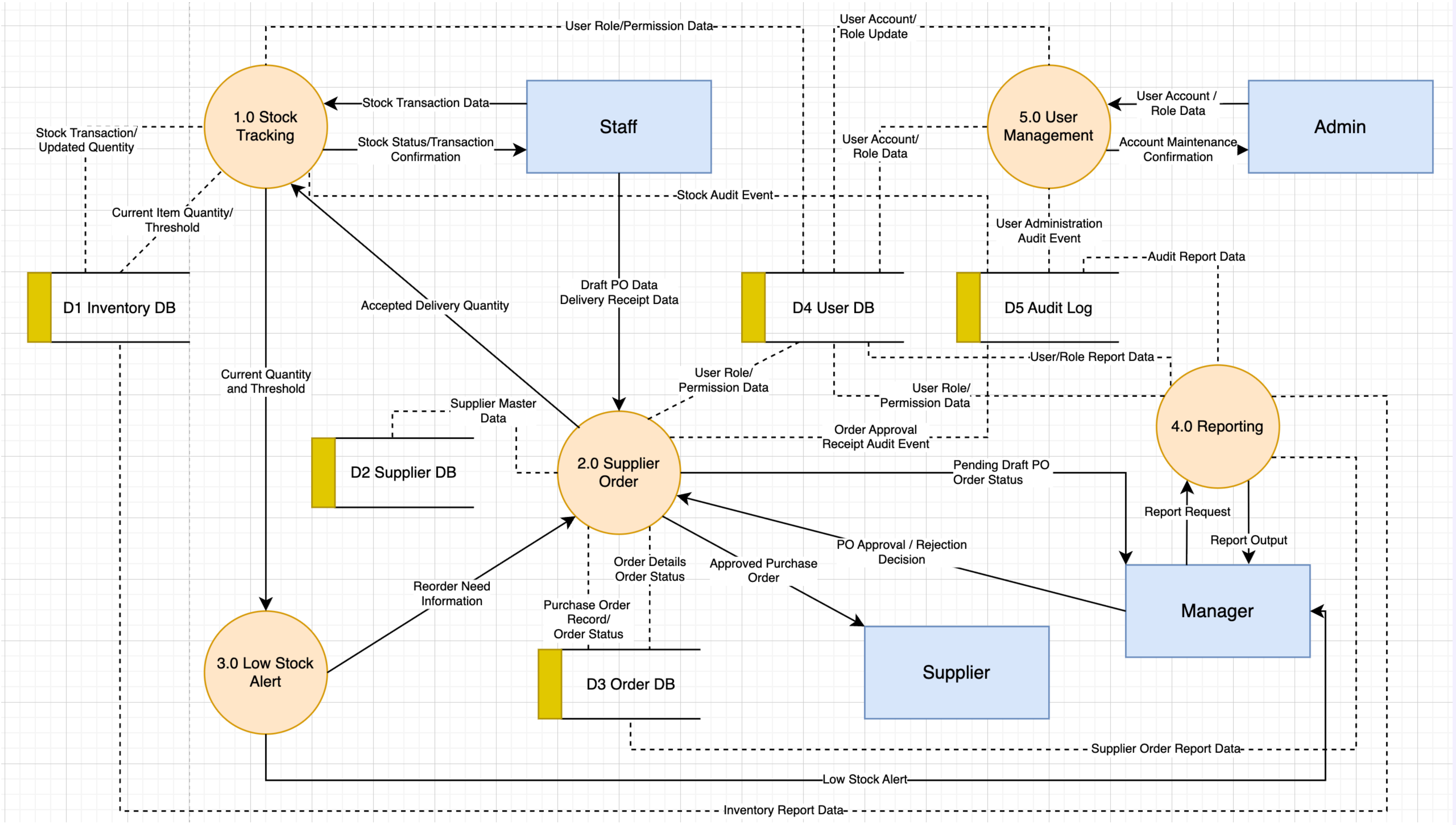
- The system shall allow Manager Users to approve or reject the draft purchase order before releasing to the supplier.

NON-FUNCTIONAL REQUIREMENTS

- The data base supports at least 100000 items, 1000 suppliers and 5 years of transaction history.

- Minimum 99% availability from 8:00 to 18:00 on working days, excluding planned maintenance.

LVL-0 DFD DIAGRAM



AMBIGUITY IN REQUIREMENTS

CONTENT

- Ambiguity occurs when requirements are unclear or can be interpreted in multiple ways.
- This makes development and testing difficult and inconsistent.

EXAMPLES

- “The system shall update stock fast” → “fast” is unclear
- “Managers can view reports when needed” → not specific
- “Supplier receives purchase order” → method not defined

IMPROVED (CLEAR REQUIREMENTS)

- System updates stock within 3 seconds
- Managers can access specific reports after login
- System generates and emails purchase order to supplier

CONFLICT (CONFLICTING REQUIREMENTS)

CONTENT

- Conflicts happen when different stakeholders have opposing needs.
- These cannot be satisfied at the same time without compromise.

EXAMPLES

- Instant alerts vs fewer interruptions
- Open access vs strict security control
- Delete old data vs keep audit history

SOLUTIONS

- Group alerts instead of sending too many
- Allow staff to create orders but require manager approval
- Use archiving instead of deletion

TESTING & QUALITY ASSURANCE

CONTENT

- Testing ensures the system works correctly and meets requirements.

TYPES OF TESTING

- Functional Testing: checks features work correctly
- Negative Testing: ensures invalid actions are rejected
- Edge Testing: checks system behavior at limits

EXAMPLES

- Add stock → quantity increases
- Invalid login → access denied
- Stock = threshold → alert triggered

What is QA Testing and Why is it Important for Software Development?

FUNCTIONAL TEST CASES

ID	Scenerio	Input	Expected Output
TC-F-01	Create stock receipt	Staff records Item A receipt of 25 units.	Inventory quantity for Item A increases by 25 and a success message is diaplayed.
TC-F-02	Issue Stock	Staff issues 5 units of Item B	Inventory quantity for Item B decreases by 5 and the transaction is saved.
TC-F-03	Create draft order	Staff creates a draft purchase order for Supplier S1 with two low-stock items.	A draft order is saved with status Draft and an order number is generated.

DEFECT REPORT

Content:

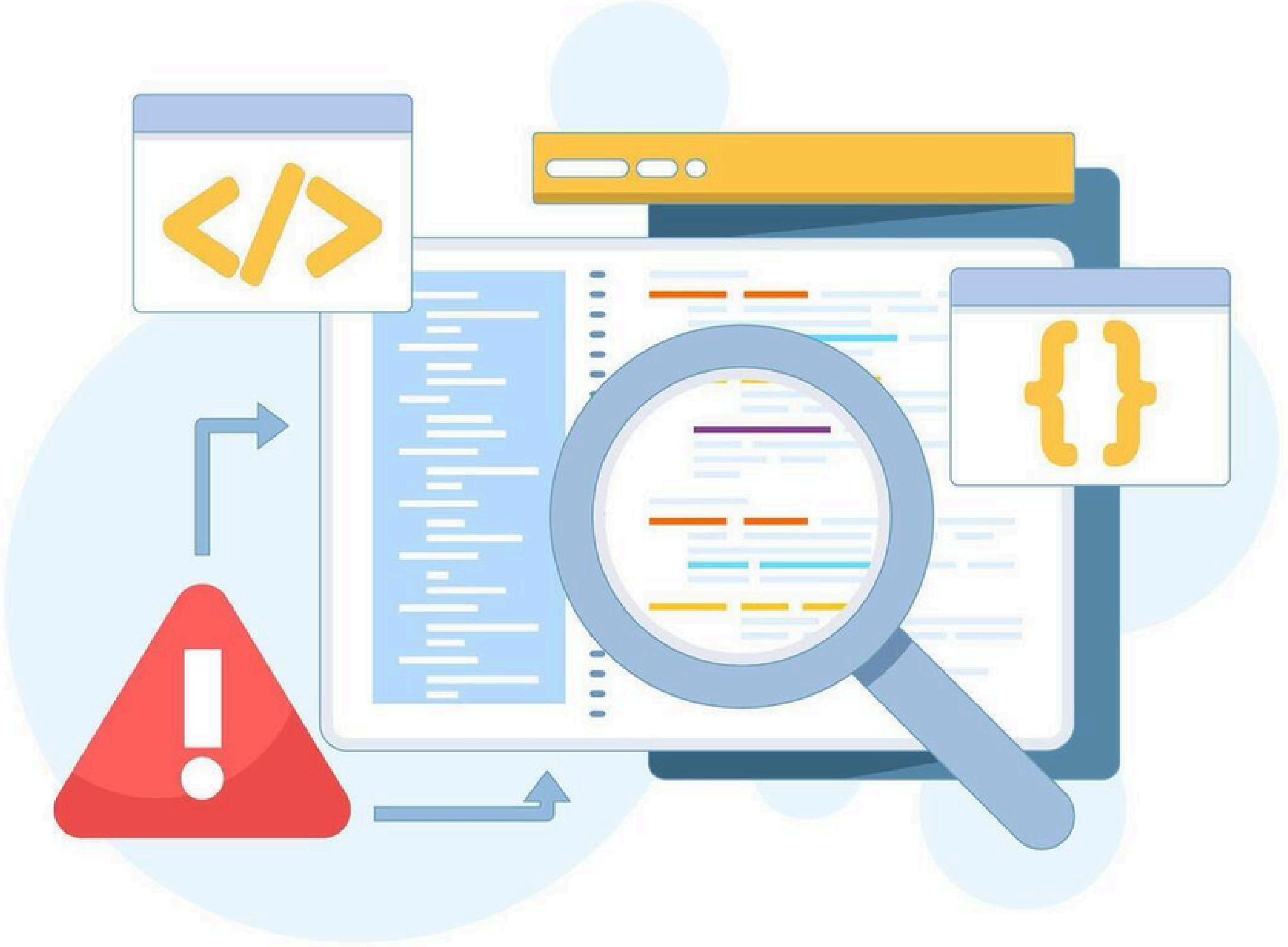
- A defect report records system errors for fixing and tracking.

Defect Example 1:

- Issue: Stock not updated after delivery
- Expected: Stock increases
- Actual: Stock unchanged
- Severity: High

Defect Example 2:

- Issue: No alert when stock = threshold
- Expected: Alert triggered
- Actual: No alert
- Severity: Medium



THANK YOU